Stakeholder Report

Dataset: heart | Best model: CatBoost | Test score: 0.8914

User Request and Task Understanding

The system predicts which patients are at risk of heart disease, prioritizing catching every real case, even at the cost of false alarms.

Core Task and Data Challenges

This is a classification task with 918 rows and 12 columns, featuring a mix of numeric and categorical data. The target variable has a low severity imbalance, with the top class accounting for 55.31% of the data. The preprocessing steps were influenced by the initial data profile, which showed that certain columns required specific handling.

EDA Findings and Preprocessing Decisions Applied

Exploratory data analysis measured the distribution and imbalance of the target variable. Preprocessing involved filling missing values using median for Age, RestingBP, Cholesterol, FastingBS, MaxHR, and Oldpeak, and mode for Sex, ChestPainType, RestingECG, ExerciseAngina, and ST_Slope. One-hot encoding was applied to Sex, ChestPainType, RestingECG, ExerciseAngina, and ST_Slope, while standard scaling was used for Age, RestingBP, Cholesterol, and MaxHR, and robust scaling for FastingBS and Oldpeak. Outliers were handled with IQR capping for FastingBS and Oldpeak, and six features were dropped due to near-zero mutual information with the outcome.

Feature Engineering

No new features were created. The recorded reason for skipping feature engineering is that the preprocessing plan flags feature_engineering_likely=False.

Model Selection and Evaluation Metric

The metric used is the weighted F1 score. The top three models are CatBoost with a test score of 0.8914, RandomForest with a test score of 0.8858, and GradientBoosting with a test score of 0.8698. The gap between the top model and the runner-up is 0.0056. Measured on 184 held-out rows. A majority-class guesser would score 0.5543 accuracy. This model scores 0.8913 (+0.3370). It catches 91 of 102 (89.2%) actual positive cases, misses 11, and raises 9 false alarms against 73 correctly cleared.

Takeaway

Features like ST_Slope_Up, Cholesterol, and Oldpeak carry significant weight in the model. The fact that these features are important suggests that the model is capturing meaningful patterns in the data related to heart disease risk. This AutoML run delivers a model that can identify at-risk patients with a high degree of accuracy, meeting the user's goal of prioritizing the detection of true cases over avoiding false alarms.